

Using ECoG Data to Test Inverse Graphics Models for Face and Body Processing

Abstract

The visual system is extraordinarily efficient at grasping its surroundings. Upon opening our eyes, we are able to rapidly dissect and interpret sensory data. The efficient inverse graphics (EIG) hypothesis, which posits that the brain reconstructs 3-D scenes from 2-D images, is one way of understanding this capacity. Past work has aligned EIG models with macaque perception of faces and bodies, but we have yet to ascertain if these findings are applicable to humans.

Our project aims to evaluate the cross-species transferability of EIG models by analyzing model predictions in conjunction with electrocorticography (ECoG) data in epilepsy patients. This new dataset records, in high spatio-temporal resolution, the cognitive responses of subjects exposed to various forms of stimuli. To account for the variability in ECoG electrode coverage across patients, we plan to establish a data-driven pipeline to identify face- and body-selective electrodes. A separate objective is to adapt or build EIG models for face and body processing. If these two goals are met, we can then evaluate the fidelity of our model predictions by applying representational similarity analysis to the distribution of ECoG signals.

It is hypothesized that EIG models can accurately capture neural behavior across the ventral visual pathway, mirroring prior results in macaque electrophysiology. This project aims not only to verify if EIG can serve as a foundation for face and body processing, but also to advance existing techniques of analysing ECoG data and inspire biology-informed models of computer vision.

Background and Significance

Perception involves inference. Our eyes take in sensory data such as shape and texture, which the visual cortex then uses to generate qualitative conclusions. This idea is foundational to vision theory and is supported by philosophers of neuroscience, who assert that inference enables us to recognize the underlying causes of stimuli and the invariant properties of objects (Brendan Ritchie, 2022). The approach of “inverse graphics” is further supported by behavioral studies. For instance, Erdogan and Jacobs (2017) designed an inference model that, compared to various alternatives, best captured the cognitive responses of subjects given a shape identification task. Thus, plentiful evidence motivates us to model perception as a set of inferential calculations.

In particular, **efficient inverse graphics (EIG)** theorizes that the visual system *inverts* the graphics pipeline, which adapts 3-D scenes to 2-D displays, in order to reconstruct the real world from flat retinal images (Yildirim et al., 2020). Due to its focus on inferring the underlying properties of a scene (in contrast to purely object classification-based models), EIG emulates the visual system’s capability to rapidly incorporate parameters including lighting, perspective, and changes in shape, enabling robust and consistent object recognition (Yildirim et al., 2024).

EIG models have two major components: a **generative graphics model** and a **deep convolutional neural network** (Yildirim et al., 2020). Depicted as **(i)** in Figure 1, the graphics model generates 2-D images when given a prompt. **(ii)**, the convolutional neural network, inverts this process step-by-step, each of its layers roughly corresponding to a dimensionality reduction step in (i), until it reconstructs the original 3-D scene.

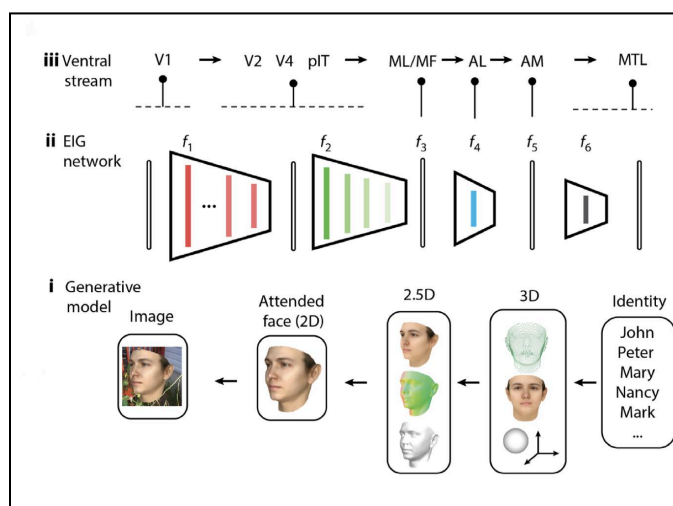


Figure 1. An EIG model for face processing (obtained from Yildirim et al., 2020). The layers of the model map onto the steps involved in macaque ventral-stream face perception.

EIG is supported by studies of macaques, whose visual systems are remarkably similar to ours (Horwitz, 2015). Of particular interest is a dedicated network of six regions in the inferotemporal (IT) cortex: the macaque “face patch system.” Applying functional magnetic resonance imaging (fMRI) and electrophysiological techniques to this network, neuroscientists were able to unravel the face processing pipeline in macaques. Moreover, they discovered “patches” specialized for perceiving color, scenes, and bodies (Hesse & Tsao, 2020).

Using these findings, Yildirim et al. (2020) designed an EIG model inspired by face patch circuitry that closely captured the cognitive behavior of macaques, registered through high-resolution single-unit neural recordings. This and related studies (into body processing, for instance) provide evidence that the EIG hypothesis holds for non-human primates. To investigate this idea in humans, we resort to an alternative neuro-imaging technique.

Electrocorticography (ECoG) is a procedure that involves placing electrodes directly on the cortical surface to capture electrical signals. It is closely related to the more popular technique of electroencephalography (EEG), which pastes electrodes onto the scalp. But by using intracranial recordings, ECoG obtains comparably higher spatial resolution (<1 cm), a high signal-to-noise ratio, and reduces its susceptibility to physiological artifacts. Moreover, it attains far finer temporal resolution (<5 ms) than fMRI (Hill et al., 2012).

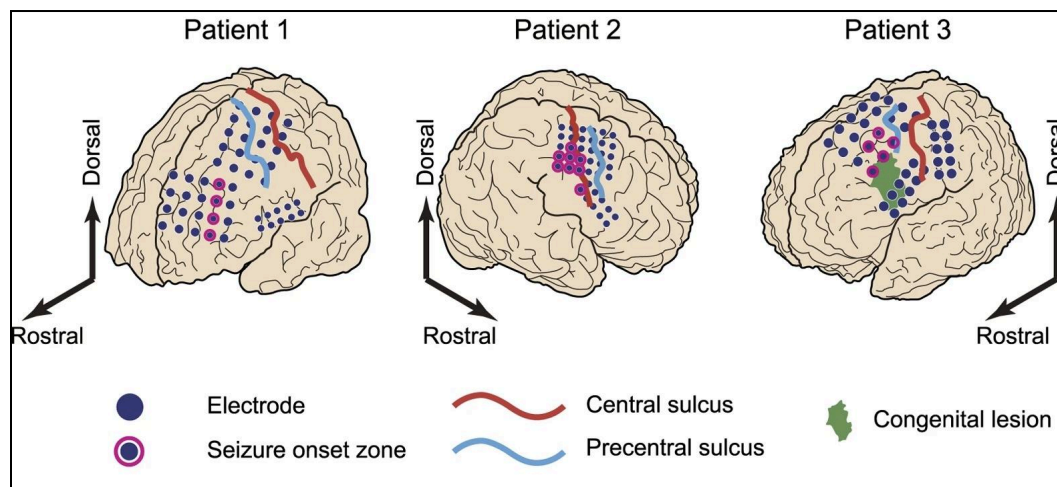


Figure 2. Potential ECoG sites (obtained from Nakanishi et al., 2017). Electrodes are placed on varying regions of the cortical surface depending on the patient’s seizure onset zone.

These advantages make ECoG suitable for studying the visual system, which involves rapid neural responses, particularly in regions of the IT cortex and ventral stream that are linked

to object recognition. It is crucial to track *when* and *where* electrical activity occurs, and this criterion is satisfied by the high spatio-temporal resolution of ECoG data. As ECoG is an invasive technique, it is mostly used in patients with intractable epilepsy to locate seizure onset zones (Hill et al., 2012). As such, existing clinical datasets, including the unpublished data we intend to use, are not standardized; they require thorough pre-processing to correct for variability in electrode coverage. This makes ECoG a tricky yet valuable tool for exploring the neural dynamics underlying key cognitive processes.

Thus, we propose to investigate the use of ECoG data in studies of high-level object representation, and particularly to clarify the extent to which EIG is a general organizing principle of face and body recognition across species.

Hypothesis

I hypothesize that visual representations (particularly of faces and bodies) in the ventral temporal cortex are converted, according to EIG principles, from 2-D images to 3-D scenes. More definitively, I expect that modeling and inverting the probabilistic generation of 2-D images emulates the behavior of face- and body-selective regions in the ventral visual pathway, as observed through the distribution of ECoG signals.

Specific Aim 1: Establish a data-driven pipeline for identifying face- and body-selective ECoG electrodes.

Rationale: As ECoG procedures are clinical, there are inevitable variations in electrode coverage based on a patient's specific seizure onset zone (Nakanishi et al., 2017). Moreover, patients in our dataset were not given identical stimulus sets. This hinders fair comparison and obstructs our goal of creating a standardized dataset of high-quality cortical activity. Hence, a structured pipeline for registering and selecting electrodes is key. By mapping electrodes onto a shared cortical surface, we hope to reveal regions of interest and establish a reliable way of comparing neural responses across individuals. This is a crucial first step in enabling the identification of guiding principles of face- and body-processing. Moreover, designing an efficient pipeline for ECoG data is a methodological advance in its own right, serving as a resource for future studies into datasets of intracranial recordings.

To identify face- and body-selective electrodes, we exploit a shared screening task given to every patient included in the ECoG dataset. (Other tasks were given on a varying basis.) This stimulus set is diverse, incorporating images of faces, bodies, objects, and abstract art, enabling us to evaluate how strongly each electrode responds to varying types of stimuli. There are a range of statistical tools that can then be applied to the distribution of ECoG signals to isolate face- and body-selective electrodes. In particular, we elucidate the concepts of **evoked potentials** and **high-gamma power**.

Evoked potentials are electrical potentials recorded after stimulus presentation, calculated as a time-locked average of ECoG activity (Laureys et al., 2009). The expected period of focus is 200–400 ms post-stimulus, and evoked potentials in this window can be used to quantify electrode response (Yildirim et al., 2020). Tracking the prevalence of certain frequency bands is also useful for this purpose. ECoG studies often focus on high-gamma power (HGP) (>80 Hz), as activity of this level is modulated not only by stimuli, but also by task-related cognitive states (Dubey & Ray, 2020). Thus, HGP is a key tool for functional mapping, and is used to identify sensorimotor, visual, and memory regions (Gruenwald et al., 2023). Using one or more of these techniques, we can locate and map electrodes of interest.

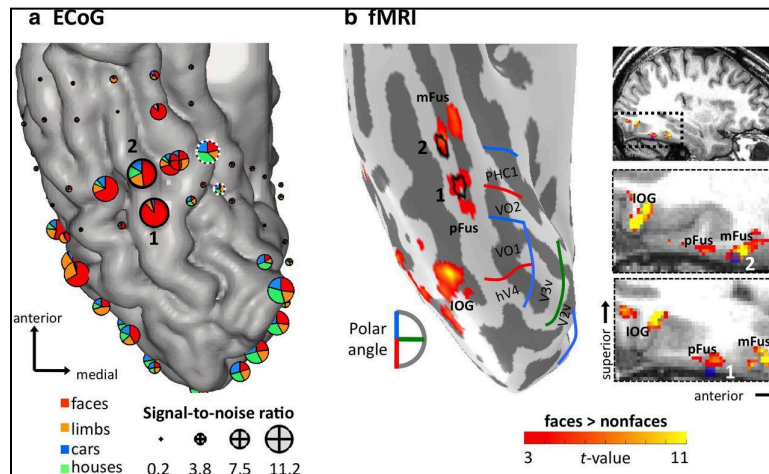


Figure 3. ECoG electrodes labeled by selectivity for four types of visual stimuli (obtained from Parvizi et al., 2012). Each pie chart depicts, for an electrode, the magnitude of response elicited by each type of stimuli. The “red” ones are face-selective, and a few are body-selective.

Next, we extend this identification procedure to group relevant electrodes into functional regions of interest. The required data is supplied by fMRI and computerized tomography (CT) scans of each individual patient. **By projecting these images onto a shared cortical surface,**

we can obtain a map of face- and body-selective electrode clusters that reveals relevant regions of the brain. Based on past work in functional mapping, we expect to find face-selective electrodes in the fusiform and occipital face areas, and body-selective electrodes in the extrastriate and fusiform body areas (Taubert et al., 2022). The activation sequence of these electrodes helps us understand *where* and *how* representations are transformed in the ventral visual pathway, providing key data for comparative analyses of EIG models.

Specific Aim 2: Develop or adapt EIG models for face and body processing.

Rationale: Advanced neural networks, in spite of first-rate accuracy in classification tasks, have evolved to share little overlap with biological vision. They rarely predict or explain psychological observations, and, therefore, are not ideal models of perception (Bowers et al., 2023). Conversely, EIG models closely fit macaque neural and behavioral data (Yildirim et al., 2020) and are able to infer contextual parameters in a human-like way (Yildirim et al., 2024). Thus, the second objective of this project is to design EIG models faithful to the neural circuitry underlying object representation in the human brain.

As stated in the Background section, EIG models have two major components: a **generative graphics model** and a **deep convolutional neural network (DCNN)**. The graphics model itself consists of multiple levels. Its top-level variable specifies an identity (a name or object) and second-level variables refer to scene parameters, including intrinsic factors of the identity (shape and texture) and extrinsic factors (lighting and perspective) (Yildirim et al., 2020). This generative component can be made by adapting external models to our specific use case. In particular, the Basel Face Model (Paysan, 2009) and Skinned Multi-Person Linear Model (Loper et al., 2023) are state-of-the-art tools for encoding 3-D faces and bodies, and can be incorporated into the architecture of our EIG models using MATLAB and Python.

The core of an EIG model is its inference network, consisting of a DCNN built on the standard ResNet architecture that exploits residual connections to enhance stability (by mitigating “vanishing gradient” issues) and performance (by expanding each layer’s access to information) (He et al., 2016). Its exact logistical details are not crucial. The key aspect is that by predicting the properties of a scene, the DCNN reverses the generative process described above (Yildirim et al., 2020). This is what differentiates EIG from classification pipelines; instead of guessing at an object’s identity, it first approximates the overall scene and only then

infers a conclusion about the object itself. By training the DCNN in a specific way, we can integrate this function into the layers of the inference network (Yildirim et al., 2020).

Thus, we have elucidated the core concepts of the EIG model and the external tools we plan to incorporate into its architecture. The construction of such models is a key step towards simulating biological perception. We hope to advance a perspective on computer vision that focuses not just on classification accuracy, but also on fidelity to cognitive processes. In service of this goal, the generative and inferential structure of EIG is inspired by our qualitative understanding of perception (Brendan Ritchie, 2022) and the ventral visual pathway (Taubert et al., 2022). **Moreover, if Aim 1 is met, we can quantitatively evaluate the correlation between EIG model predictions and ECoG data.**

To do so, we make use of **representational similarity analysis (RSA)**, a multivariate technique that enables us to unite disparate types of data. RSA characterizes the *representational space* of a stimulus set by calculating pairwise distances between stimuli, and visualizes this result as a **representational similarity matrix (RSM)** (Popal et al., 2019). By creating and comparing RSMs for neural data (pairwise distances in time-frequency features for each face and body stimulus) and model predictions (pairwise distances in the model's outputs for the same stimulus set), we can test our EIG model against the ECoG dataset.

Lastly, by contrasting the correlation between RSMs for face and body stimulus sets, we can evaluate if there is a significant gap in how body parts are perceived, as suggested by Taubert et al. (2022). That is, we can quantify the extent to which face processing is a unique domain.

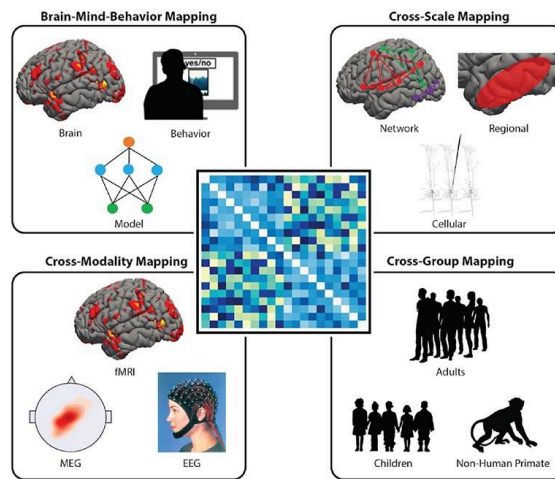


Figure 4. A schematic depicting ways that RSA can connect various data sources (obtained from Popal et al., 2019), including cross-modality and mind-model mappings. At its center is an RSM.

Potential Pitfalls and Alternative Strategies

One possible obstacle is the **variability in electrode coverage**. Many patients may not have electrodes placed in face- or body-selective regions (e.g. the fusiform gyrus and occipital lobe), which could affect the sample size and consistency of our findings. If this turns out to be an issue, we could supplement initial group-level analyses (of patients with sufficient electrode coverage) with individual analyses to capture the influence of such variations. Moreover, it is not only the canonical “face patch” and “body patch” systems that respond to these types of stimuli. There is evidence that parts of the anterior temporal lobe (ATL) and superior temporal sulcus (STL) are face-selective, despite not belonging to the core group outlined in Aim 1. Secondary body-selective regions are also found in the ATL and STL (Taubert et al., 2022). Tracking electrodes in these regions that consistently respond to face and body stimuli can yield valuable insights, even though they may not align with the established tenets of the visual pathway.

Another potential pitfall involves **issues with our EIG model**, such that it is unable to accurately capture the visual system’s inferential calculations. This could be caused by a variety of factors, but is particularly likely if the model’s understanding of scene parameters (e.g. lighting and perspective) differ from our perceptual biases: a major issue for CNNs, though predicted to be less of one for EIG models (Bowers et al., 2023). If this occurs, we would try to incorporate more advanced neural network architectures that are designed to emulate primate behavior (Dapello et al., 2023). If the issue persists, we could resort to the use of simpler feed-forward CNNs (taking away the inverse component) to see which model fits best with the empirical observations obtained from our ECoG dataset.

By preparing backup plans for our initial goals, we expect that even if we encounter unexpected obstacles, this project is likely to yield meaningful insights into the guiding principles of high-level object representation.

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